ASTANA IT UNIVERSITY

«Approved»
by the Dean
__________ Syzdykova Z.A.
«___» of __________ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	**Calculus I**
Degree cycle (level)/ major	Bachelor, Software Engineering
Year, term	1, 1
Number of credits	5
Language of delivery:	English
Prerequisites	School math courses
Postrequisites	Calculus II, Theory of Probability and Statistics
Lecturer(s)	Raikhan Madi, Candidate of Physical and Mathematical Sciences, Associate Professor, madi.raikhan@astanait.edu.kz Astana IT University, Expo, C1 block, 2nd floor, office C1.3.249
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	Calculus is a foundational course at Astana IT University; it plays an important role in understanding science, engineering, economics, and computer science, among other disciplines. This Calculus-1 course covers the differentiation and integration of functions of one variable with applications.
2. Course Goal(s)	After completing this course, students should have developed a clear understanding of the fundamental concepts of single-variable calculus and a range of skills that allow them to work effectively with them. The main goal is to be able to apply the tools mentioned above to problems in post-requisite courses.
3. Course Objectives:	Students should study the following concepts: • Sets and functions of one variable • Limits of sequences and their properties • Continuity and properties of continuous functions • Derivatives and their applications • Antiderivatives and methods to evaluate them • Definite integrals in 1D and 2D cases • Improper integrals
4. Skills & Competences	After completing this course, students should demonstrate competency in the following skills: • Define the domain and range of the elementary functions • Use both the limit definition and rules of differentiation to differentiate functions. • Use L'Hospital's rule to evaluate certain indefinite forms. • Sketch the graph of a function using asymptotes, critical points, the derivative test for increasing/decreasing functions, and concavity.

	• Apply differentiation to solve applied max/min problems. • Evaluate integrals both by using the Fundamental Theorem of Calculus. • Evaluate integrals using integration techniques, such as substitutions and integration by parts. • Apply integration to compute arc lengths, volumes of revolution, and surface areas. • Determine convergence/divergence of improper integrals and evaluate convergent improper integrals.
5. Course Learning Outcomes:	By the end of this course, students will be able • to understand concepts related to sets, limits, continuity, derivatives, and basic integrals; • to apply and work with these concepts numerically, graphically, and analytically; • apply the tools mentioned above to problems in post-requisite courses.
6. Methods of Assessment	• Homework • Classwork assignments • Quiz • Midterm • Pre-Final and Final Exams
7. Reading List	Assigned reading materials and presentations should be read before the class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and / or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). Basic Literature: 1.Lecture presentations; 2. Thomas' Calculus. By George b. Thomas, revised by J.Hass, C.Heil. M.D.Weir, Pearson Publishing Company. 14n edition 3. George b. Thomas, Jr., Ross L. Finney, Calculus and Analytic Geometry. Part II. Addison-Wesley Publishing Company. 9 th edition. Supplementary literature: 4. G. N. Berman, A collection of problems on a course of Mathematical Analysis 5. Г.М.Фихтенгольц. Основы математического анализа, Т.1, Изд-е 9-ое, Изд. Лань – 2008. – 448 с. 6. Ибрашев Х.И., Еркеғұлов Ш.Т. Математикалық анализ курсы. - Алматы, 1970. 7. Темірғалиев Н.Т. Математикалық анализ. 1 бөлім. - Алматы: Мектеп, 1987.
1. Resources	Online journals, articles, papers, books, and internet resources, as well as online emulators and online software for simulation.
2. **Course policy**	**Course and University policies include:** **Attendance:** Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to

	achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course. In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing. **Preparation for Class**: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties: - Offers a different and unique, but relevant, perspective; - Contributes to moving the discussion and analysis forward; - Builds on other comments. **Class work**: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes. **Being late on class:** When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor. **Attestation I and II:** Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course. **Home work / Assignments**: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be

	properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures. **Late submissions:** Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale. *In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.* **Final exam* / Final project:** At the completion of this course, the Final Exam will be given in the written form **Laptops and mobile devices** can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. **Online lessons** can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. **Cheating and plagiarism** are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. **Academic Conduct Policies of the university:** The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). **Contacting the Course instructor**: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	PA	Practical assignment
4	MCQ	Multiple choice quiz

3.1 Lecture, Practical Sessions Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	**Sets and Functions.** Sets and Operations on Sets. Functions and their graphs. Shifts, and Scaling graphs. Elementary functions, the transcendental functions. Inverse Functions. The composition of functions.	3	2	1	9
2	**Sequences and Limits.** Sequences converging to a limit. Properties of convergent sequences. Monotone sequences. Limits of function values. Definition of a Limit. The Sandwich theorem.	3	2	1	9
3	**Limits and Continuity.** Continuous functions. One-Sided limits. Limits involving infinity. Relative Rates of growth. (Table of equivalent functions).	3	2	1	9
4	**Derivatives**. The definition of derivative. Rules. Derivatives of trigonometric and transcendental functions. The Chain rule. Second and higher order derivatives. Implicit differentiation. Linearization and differentials. Derivative of the parametric functions, Tangents and normal of lines.	3	2	1	9
5	**Application of derivatives**. Maxima and Minima. The mean value theorem. Graphing by the first and the second derivatives. Asymptotes and Dominant terms. Graphing Rational Functions. Related rates of change. Optimization. Indeterminate forms and L'Hopital's Rule theorem	3	1	1	9
5	**Midterm Exam**		1		

6	**Integrals.** Antiderivatives. Indefinite integrals. Basic Integration Formulas. Substitutions in indefinite integrals.	3	2	1	9
7	**Integrals.** Integration by part. Special cases of integration. Integrals for Trigonometric and Rational Functions, Partial Fractions.	3	2	1	9
8	**Definite integrals.** Area and estimating with finite sums. Sigma notation and limits of finite sums. The fundamental theorem of Calculus. Substitution in definite integrals. Improper integral.	3	2	1	9
9	**Application of Integrals.** The area between curves. Volumes. Arc Length. Areas of surfaces of revolution.	3	2	1	9
10	**Complex numbers.** Complex numbers and operations with them. **Polar coordinates**. Graphing polar coordinate equations. Integration in polar coordinates. Area and Lengths in polar coordinates.	3	1	1	9
10	**End-term Exam**		1		
	Total hours:	**30**	**20**	**10**	**90**

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	**2**	**3**	**4**	**5**
1	1.1 Functions, their domain, range, and graphs. Shifts, and Scaling graphs. Inverse Functions. The composition of functions.	9	[2]: Ch. 1.1 – 1.3 and Ch. 7.1-7.3, Appendix A.1	Electronic version or hard cory of the assignment
2	1.2 Sequences converging to a limit. Limits of convergent sequences and functions by definition.	9	[2]: Ch. 8, Appendix 10.1, [3], Ch1	Electronic version or hard cory of the assignment
3	2.1 Computing Limits of Function values and applying Theorems. Limits involving infinity..Estimate relative Rates of growth.	9	[2]: Ch. 2 and 7	Electronic version or hard cory of the assignment
4	2.2 Compute derivatives by definition and applying rules. Derivatives of trigonometric and transcendental	9	[2]: Ch. 3 and 7	Electronic version or hard cory of the assignment

	functions. The Chain rule. Second and higher order derivatives. Implicit differentiation. Linearization and differentials. Derivative of the parametric functions, find tangents and normal of lines.			
5	3.1 Find Maxima and Minima. Graphing by the first and the second derivatives. Find Asymptotes and Dominant terms. Apply L'Hopital's Rule and compute limits for Indeterminate forms	9	[2]: Ch. 4,7	Electronic version or hard cory of the assignment
6	3.2 Compute antiderivatives. Use Basic Integration Formulas. Make Substitutions in indefinite integrals	9	[2]: Ch. 5, 8	Electronic version or hard cory of the assignment
7	3.3 Apply integration by part. Study special cases of integration for Rational and trigonometric Functions	9	[2]: Ch. 5, 8	Electronic version or hard cory of the assignment
8	4.1 Compute the area between curves and Areas of surfaces of revolution.	9	[2]: Ch.5, 8	Electronic version or hard cory of the assignment
9	4.2 Compute volumes and Arc Length.	9	[2] Ch. 6, 8	Electronic version or hard cory of the assignment
10	4.3 Exercise with complex numbers and polar coordinates	9	[2]: Ch. 11.3 – 11.5, Appendix A.7	Electronic version or hard cory of the assignment

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1st attestation	**Assignments**:** assignment 1, assignment 2 Quiz 1 **Mid Term**	**60** 20 20 20 **40**	100
2nd attestation	**Assignments**:** assignment 3, assignment 4, Quiz 2 **End Term**	**60** 20 20 20 **40**	100
Final exam*	Written Exam		100
Total	**0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final**		**100**

** *The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

* *Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1.67	60-64	
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	Fail
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and

	material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge.
- **Final assessment** is a Written Exam.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Raikhan Madi	Associate professor	25/08/2024	

Director **Sergaziyev M.Zh.**